
CONCEPT SUMMARY: Find Sums and Differences of Rational Expressions

WORDS

To add or subtract rational expressions with common denominators, add the numerators and keep the denominator the same.

To add or subtract rational expressions with different denominators, rewrite each expression so that its denominator is the LCD, then add or subtract the numerators.

NUMBERS

$$\frac{1}{5} + \frac{3}{5} = \frac{1+3}{5} = \frac{4}{5}$$

$$\frac{1}{6} + \frac{1}{15} = \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 5} = \frac{1 \cdot 5 + 1 \cdot 2}{2 \cdot 3 \cdot 5}$$

ALGEBRA

$$\frac{x}{x+4} + \frac{5}{x+4} = \frac{x+5}{x+4}$$

$$\begin{aligned} \frac{x+3}{x^2-1} + \frac{2}{x^2-3x+2} &= \frac{x+3}{(x+1)(x-1)} + \frac{2}{(x-1)(x-2)} \\ &= \frac{(x+3)(x-2)}{(x+1)(x-1)(x-2)} + \frac{2(x+1)}{(x+1)(x-1)(x-2)} \end{aligned}$$
